

Qubits, Bloch Sphere (Lecture 1)

Course: Survey of Quantum Technologies & Applications (PH 40001)

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What is a Qubit?

- A qubit is a quantum system in a two-dimensional complex vector space.
- Like a classical bit, it has two distinguished states: 0 and 1.
- In quantum mechanics, the qubit's states are written using Dirac notation as **kets**.

 $|0\rangle$ $|1\rangle$

Basis Vectors

- The states $|0\rangle$ and $|1\rangle$ are not just values.
- They are the basis vectors of a two-dimensional complex vector space.

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Why Basis Vectors?

- Every state in the qubit space can be written as a linear combination of the two vectors, $|0\rangle$ and $|1\rangle$.

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Superposition

- The state of a qubit can be a combination of the two computational basis states, $|0\rangle$ and $|1\rangle$.
- We call such state a superposition of $|0\rangle$ and $|1\rangle$.

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

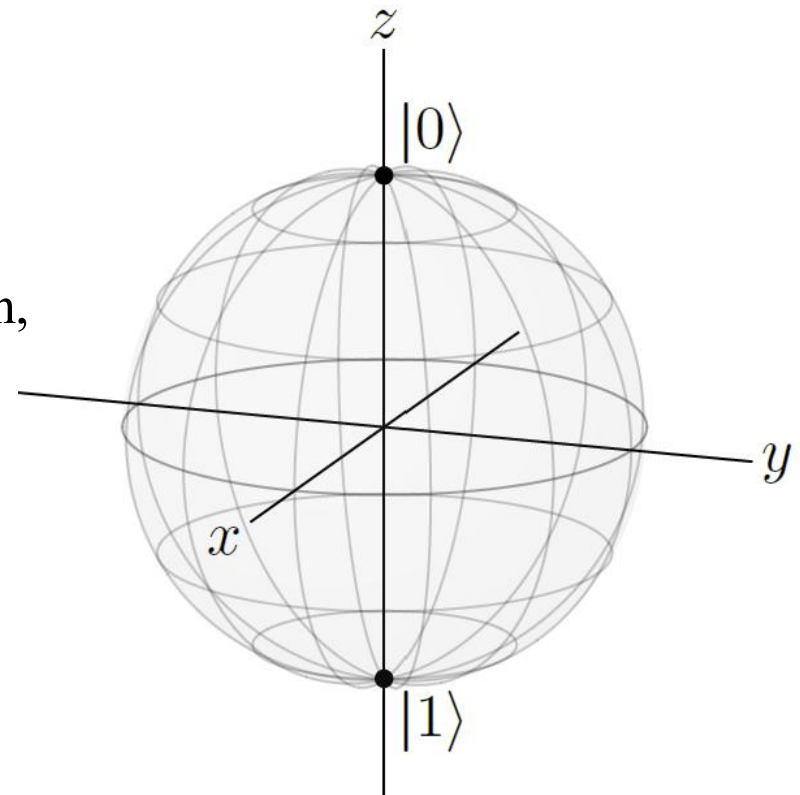
- The complex numbers α and β are called probability amplitudes.

$$|\alpha|^2 + |\beta|^2 = 1$$

Bloch Sphere

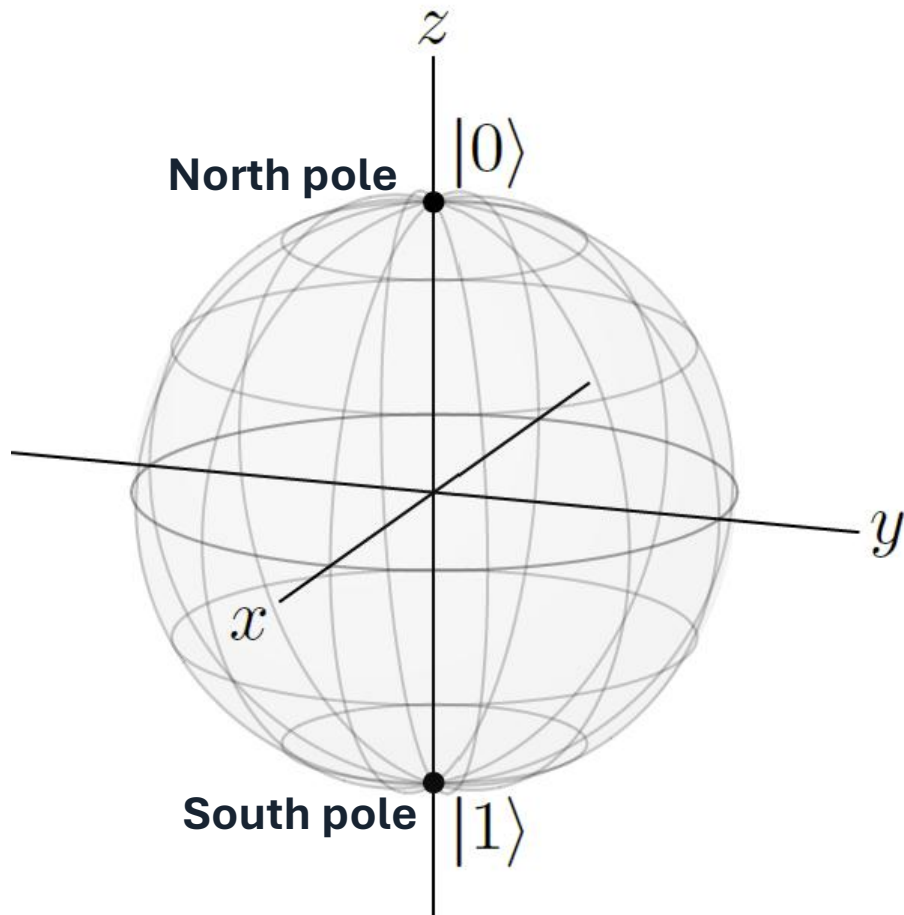
- A graphical representation of a state of a qubit

Following the standard physics convention, the x-axis comes out of the page, the y-axis points to the side, and z-axis is oriented up.



Basis States on Bloch Sphere

$|0\rangle$ and $|1\rangle$ are placed at opposite poles.



Because the radius is 1

$$|0\rangle \leftrightarrow (0,0,1)$$

$$|1\rangle \leftrightarrow (0,0,-1)$$

Superposition on Bloch Sphere

A parameterize one qubit state

$$|\Psi(\theta, \phi)\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$$

The angles θ and ϕ provide a geometric representation on the Bloch sphere.

Ranges of θ and ϕ

$$0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi$$

$e^{i\phi}$ is known as a global phase.

Superposition on Bloch Sphere

Example 1: Assume a superposition state is $\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$

How do we represent this in a Bloch Sphere?

1. Start from the standard qubit form

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

Meaning of the angles

- theta is the polar angle measured from the positive z-axis.
- phi is the azimuthal angle in the x-y plane, measured from the positive x-axis.
- The coefficient of $|0\rangle$ fixes theta; the relative phase between the amplitudes fixes phi.

Compare this form term-by-term with the given qubit state.

2. Match the amplitudes

Given state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

Therefore:

$$\cos\left(\frac{\theta}{2}\right) = \frac{1}{\sqrt{2}}, \quad e^{i\phi}\sin\left(\frac{\theta}{2}\right) = \frac{1}{\sqrt{2}}$$

Key observation

Both amplitudes have magnitude $1/\sqrt{2}$, and their relative phase is zero.

3. Calculate the polar angle theta

$$\frac{\theta}{2} = \frac{\pi}{4} \implies \theta = \frac{\pi}{2} = 90^\circ$$

4. Calculate the azimuthal angle phi

$$e^{i\phi} = 1 \implies \phi = 0$$

5. Convert to Bloch-vector coordinates

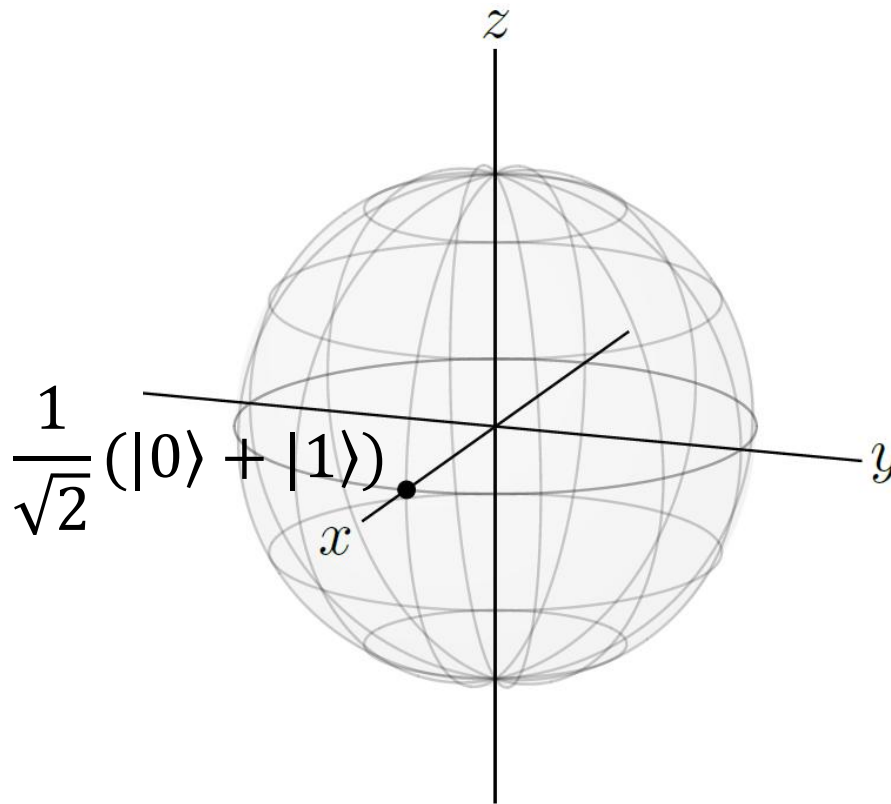
$$x = \sin \theta \cos \phi, \quad y = \sin \theta \sin \phi, \quad z = \cos \theta$$

Substitute $\theta = \pi/2$ and $\phi = 0$

- $x = \sin(\pi/2) \cos(0) = 1$
- $y = \sin(\pi/2) \sin(0) = 0$
- $z = \cos(\pi/2) = 0$

$$(x, y, z) = (1, 0, 0)$$

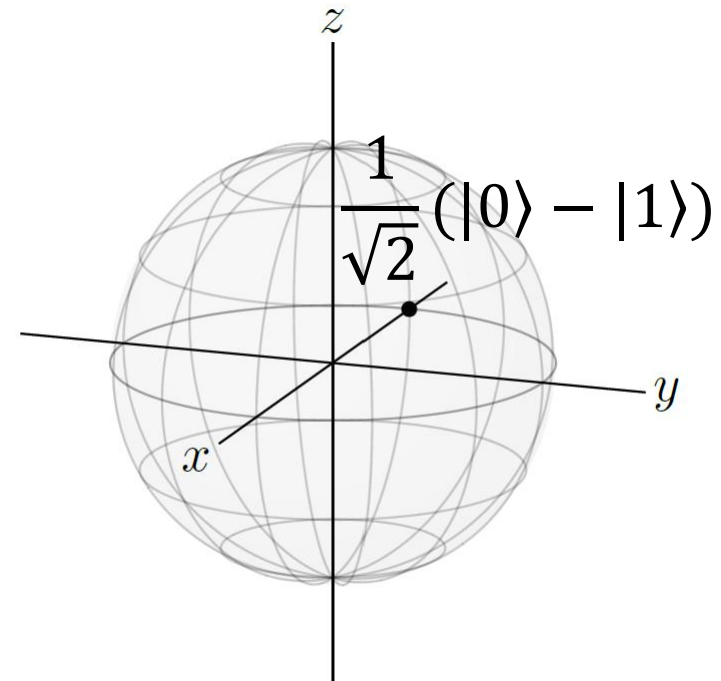
6. Draw the state on the sphere



Superposition on Bloch Sphere

Example 2: Assume a superposition state is $\frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$

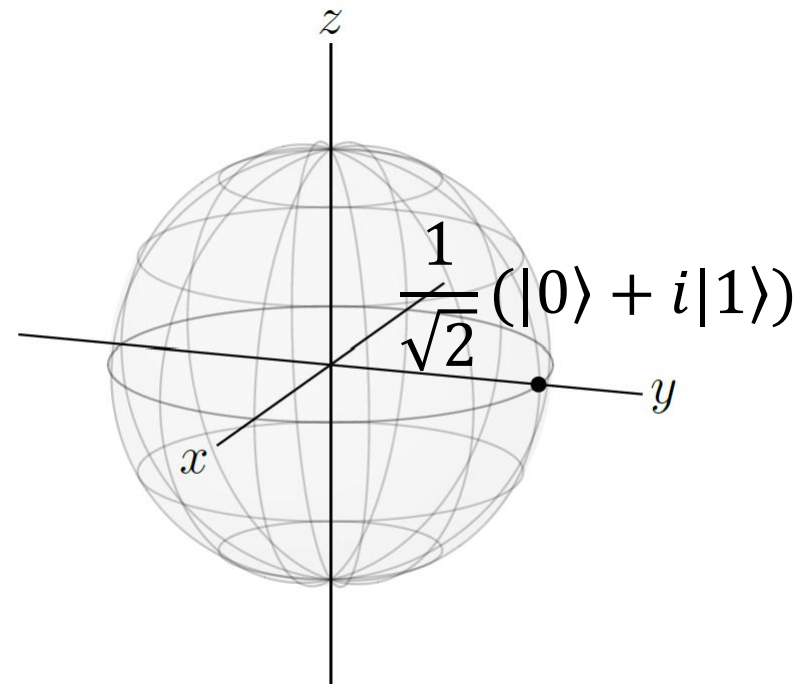
This state is at the point $(-1, 0, 0)$,
on the negative x-axis.



Superposition on Bloch Sphere

Example 3: Assume a superposition state is $\frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle)$

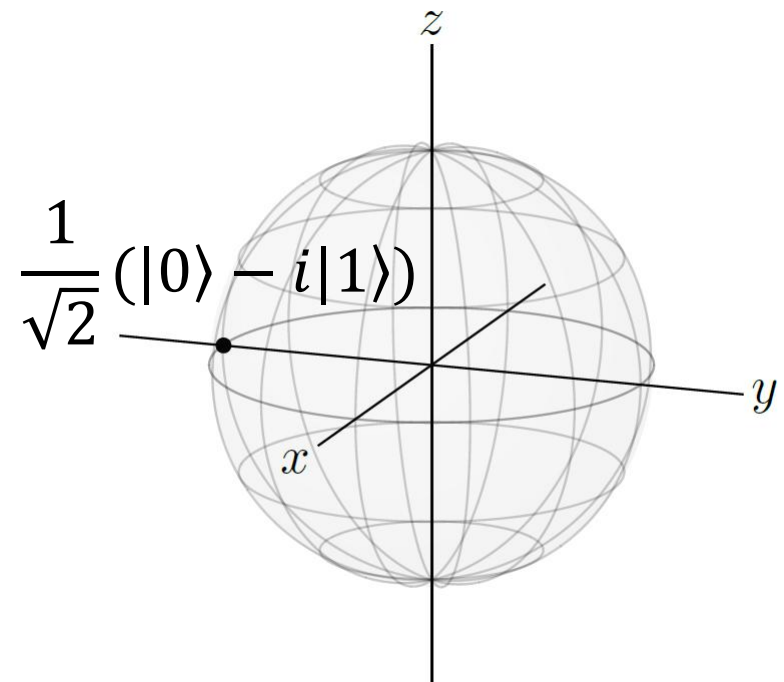
This state is at the point $(0,1,0)$,
on the positive y-axis.



Superposition on Bloch Sphere

Example 4: Assume a superposition state is $\frac{1}{\sqrt{2}} (|0\rangle - i|1\rangle)$

This state is at the point $(0, -1, 0)$,
on the negative y-axis.



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